Prime-degree comparison and published tool

2026-10-16 UTC · GOAL-ISOU-f30a07 · documentation task T-3286

The goal pursues independently verified separable prime degrees on all registered standard prime-field source models, ranked separately. This round checks primary-source comparison boundaries and refreshes installation instructions. Four metadata/help commands ran; scientific trials and executable custody uploads both remain zero.

Prime-degree comparisons require the same output contract

2026-10-16 · lower-bound literature audit · no external instance replay in this round

Expanded map + monic kernel

54 exact standard-source boards; fresh full algebra and subgroup checks

Existing source-specific incumbents remain; larger-map replay still censored.

Supplied-kernel or ideal-based point evaluator

Jao–Soukharev; square-root Vélu; parallel square-root Vélu

Published evaluations are lower bounds in their categories; expanded coefficients need checking.

Compact representation + theorem and independent evaluator

Known-CM self-maps and higher-dimensional representations stay separately identified.

PRISM prime-degree variant: numeric instance extraction and replay remain open.

No cross-category degree ranking; no new score, speed ratio or worldwide-record statement.

Existing campaign evidence

The historical pinned expanded-map incumbents remain P192/10453, P224/1471 and P521/7. A new release season requires fresh replay; old scores are not imported automatically. All 54 standard-source boards remain visible: 51 unscored, 47 construction/replay adapters, seven conversion gaps. Adapter coverage does not establish an exhaustive search.

The P192/50341 artifact was constructed, but traced replay RUN-ISOU-93b2b3 ended CENSORED/TIMEOUT with no score. Selected torsion ended; subgroup modular composition began without an end or a terminal mathematical verdict. Its frozen input and raw resource receipts remain unchanged. No nonexistence conclusion follows.

The separate known-CM compact result on exact secp256k1 has prime degree 2^44497−1. EV-ISOU-263837 records a 63615-byte certificate and independent endpoint replay under the reviewed norm theorem and trusted SEC2 p/n primality. This remains a separate category; no expanded-map score or worldwide record is established.

Frozen traced replay observation · Frozen compact evidence

Primary-source comparison boundaries

The dated Markdown report and sources.json retain the search queries, retrieval outcomes, versions and sections. This is a lower-bound survey; no external program was built or run and no external calculation was independently reproduced.

Primary source	Observed scope
Jao–Soukharev (2010), §§5.1–5.3	Degree-5000000029 and ECCp-109 point evaluations are supplied. ECCp-239 evaluation is left unfinished after an ideal relation. These examples do not supply expanded degree-sized maps.
Bernstein et al. (2020), introduction	The supplied-kernel method computes quotient curves and point images. The paper distinguishes this contract from producing all rational-map coefficients.
Chávez-Saab et al. (2024), §4	The author preprint evaluates construction plus two point images over a chosen 1792-bit prime field, for degrees spanning 19–321193. The MDPI endpoint returned HTTP429 in this round.
PRISM journal extension (2026), §§3, 5, 5.2, 7	The basic response has degree q(2^a−q). Section5 separately gives prime degree $2^{(2e-1)}-s^2$ represented in dimension two, with a four-dimensional alternative in §5.2. Section7 and the published C repository focus on the salted variant.

The earlier phrase “higher-dimensional” for the PRISM prime-degree variant is made precise: its stated construction uses dimension two. A concrete numeric instance, certified degree, exact curve, representation and independently checked evaluation are still needed before an instance-level record comparison. Neither a parameter bound nor a prime factor is an observed maximum prime degree.

The largest comparable published expanded map and global maximum evaluated degree remain unresolved. External evaluations already exceed50341 in their own categories. An accepted P192/50341 map would advance its source-specific expanded-map frontier; the worldwide comparison would remain separate.

Primary references

Jao–Soukharev, arXiv1002.4228v2
Bernstein et al., arXiv2003.10118v1
Parallel square-root Vélu, preprint202401.1366v1
PRISM, Journal of Cryptology39, article30, 2026
Authors’ salted-variant C repository; not executed

Official standalone release and remaining work

Release v0.1.2390 is published from source `bd7eb117f094a336d3e3e8f31facb0c6b86f95bc`. The preceding release QA retained exact asset hashes, successful Release jobs and native public-map admission controls. The canonical guide now gives the pinned Apple-silicon download, checksum-before-extraction instructions and all four platform filenames.

This follow-up ran `--version`, `curves`, `challenge policy` and `--help` using the downloaded Apple executable from a fresh cwd outside the checkout. All four exited0. It reports sourcebd7eb117, dirty=false, 2112 catalogue models and54 standard-prime challenge boards. Runtime package version0.1.0 is distinct from the rolling repository release tag.

The executable SHA256 is `1514b1a7d7248f716e810169218dfdbdda941d454934c93e81714d68b8e66d9e`. Complete raw output is in usage-check.tar.gz. usage-manifest.json binds commands, cwd, binary identity, per-file hashes and archive verification. usage-check.sh is a metadata-only reproduction recipe. No timing or new-degree inference follows.

Requested work	Current evidence or remaining gate
Dedicated published executable	Verified official assets and Apple native controls; installation guide refreshed.
Standard-curve coverage	54 separate boards retained; seven adapters and unsearched models remain explicit.
Larger expanded degree	50341 replay censored; no replacement scientific run in this round.
Faster replay	Default-off implementation and bounded controls committed in PR1991; measured improvement unestablished.
Matched-build custody/review	Pending authorization for the four research executables' S3 transfer. Source gate remains incomplete.
Worldwide record	Comparable expanded-map register and instance-level compact comparisons remain incomplete.

Validation, exact sources and graph status

No native source or frozen season changed. The previous root/native/research gates apply to the unchanged code closure; no example or Python implementation changed. The standalone metadata check and archive hash replay passed. All raw receipts remain available. The stored S3 approval question is pending; no rejected upload was retried.

README.md gives full ICV1 source IDs for P192, P224, P521 and secp256k1 and immutable links to current evidence. The challenge leaderboard/rules, curve registry, IC progress timeline, IC scoreboard and performance-gains sources were checked. No new verified edge, score, IC result or speed ratio arose, so their data remain unchanged. The diagram classifies comparison contracts and makes no cross-category numerical ranking.

The initial campaign/challenge packet bodies remain historical records beneath dated current pointers. A sparse-checkout command initially rejected an unsupported option; the corrected exact-path addition succeeded without tracked source changes. The next scientific step still depends on the frozen-source custody/review gate and its subsequent admitted measurement/replay protocol.